

**Table 1: Few-shot only** (different prompts and lengths than the zero-shot table—do not mix rows). Falcon-Mamba BABILong ablation results. Accuracy is exact-match (%) averaged over QA1–QA3 (50 per cell). *No cache, few-shot:* eval\_babilongv2 paper few-shot prompts (--few-shot-examples 2), candidate answer-token scoring (--babilong-scoring next\_token); metrics/fewshot\_babilong/no\_cache\_baseline\_next\_token\_qa1\_qa3.json. *StateCache rows:* cache\_acc from metrics/fewshot\_babilong/{run\_id}.json (eval\_babilong\_paper\_scaffold\_checkpoint, paper-style few-shot + [CACHE]).

| Method                                       | 0k          | 1k          | 2k          | 4k          | 8k          | 16k         | Avg.        |
|----------------------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| No cache, few-shot                           | 49.3        | 42.0        | 29.3        | 29.3        | 25.3        | 20.0        | 32.6        |
| Loss-selected, $k = 1$ , L62                 | 28.0        | 33.3        | 34.0        | 16.7        | 18.7        | 18.0        | 24.8        |
| Loss-selected, $k = 4$ , L62                 | 29.3        | 34.7        | 36.7        | 14.7        | 22.7        | 18.7        | 26.1        |
| Loss-selected, $k = 8$ , L62                 | 27.3        | 40.0        | 37.3        | 18.7        | 17.3        | 20.7        | 26.9        |
| Loss-selected, $k = 16$ , L62                | 28.7        | 33.3        | 34.0        | 12.0        | 20.7        | 14.0        | 23.8        |
| Loss-selected, $k = 32$ , L62                | 32.0        | 40.0        | 42.0        | 18.0        | 17.3        | 20.7        | 28.3        |
| Loss-selected, $k = 64$ , L62                | 31.3        | 34.0        | 38.7        | 21.3        | 18.7        | 16.7        | 26.8        |
| Loss-selected, $k = 4$ , L62, train up to 8k | 25.3        | 30.7        | 30.0        | 34.0        | <b>33.3</b> | 24.0        | 29.6        |
| Loss-selected, $k = 4$ , L62, no gate        | 32.0        | 36.0        | 37.3        | 20.7        | 18.7        | 24.0        | 28.1        |
| Loss-selected, $k = 4$ , L62, full loss      | 27.3        | 18.0        | 15.3        | 15.3        | 15.3        | 16.0        | 17.9        |
| Loss-selected, $k = 4$ , L32                 | 48.7        | 43.3        | 36.7        | 27.3        | 30.7        | 26.0        | 35.4        |
| <b>Loss-selected, <math>k = 4</math>, L2</b> | <b>64.7</b> | <b>54.7</b> | <b>51.3</b> | <b>45.3</b> | 31.3        | <b>28.0</b> | <b>45.9</b> |
| Uniform interval, $k = 4$ , L62              | 31.3        | 31.3        | 38.7        | 19.3        | 22.0        | 18.0        | 26.8        |
| Random, $k = 4$ , L62                        | 27.3        | 21.3        | 21.3        | 18.0        | 22.0        | 13.3        | 20.6        |
| Regex/entity, L62                            | 28.7        | 28.7        | 36.0        | 18.0        | 13.3        | 21.3        | 24.3        |